Course Handout (Project-I)
(Batch: 2024-2028) - (Odd Sem - 2026)

Course Code: PRJ3908	**Course Name:** Major Project-I	
Credits: 2 (2 L – 0 D – 0 P)	**Contact hours per week:** Four Hours	
Batch: 2024-2028, 5th Sem **Academic Year:** 2026-27	**Semester Duration: 27th** July 2026 to 30th Nov 2026	
Mentor In-charge: As per the student-mentor allocation.	**Course Coordinator:** Dr. Shraddha Arora Email : shraddha.arora@bmu.edu.in Contact : +91-9811352222 Address : Dr. Sukhandeep Kaur Email : sukhandeep.kaur@bmu.edu.in Contact : +91-9501880466 Address : #3, IV Floor, E2 Building	

Aim of the course:

The aim of Project I, a 2-credit course, is to serve as the pinnacle of study in pursuit of a Bachelor of Engineering degree. This course provides students with the opportunity to practically apply and expand upon the knowledge acquired throughout their academic program. The primary focus lies on fostering student development in technical proficiency, project management, and presentation abilities. This course challenges students to conduct thorough investigations within their chosen area, encompassing problem understanding, identification of gaps in existing research or methodologies, analysis of major challenges, and consideration of applicability. Students will utilize various tools, libraries, and packages to develop programs or solutions, thereby gaining valuable hands-on experience.

Course Overview and Context:

1. Perform a literature search to review current knowledge and developments in the chosen technical area.
2. Undertake detailed technical work in the chosen area using one or more of:
   - Product Design
   - Computer simulations/Hardware design/ML Tools
   - Web/Mobile app design
3. Produce progress reports or maintain a professional journal to establish complete work, and to schedule additional work within the time frame specified for the project.
4. Deliver a detailed Presentation on the area of work being undertaken and specific contributions to that field.
5. Prepare a formal report describing the work undertaken and results obtained so far; and
6. Present the work in a forum involving poster presentations and demonstrations of operational hardware and software.

Course Outcomes (COs):

At the end of the course the students should be able to do the following:

	Course Outcome Statement	Mapped Component	Associated Activities / Assessment
CO1	Understand computer science engineering principles to identify, formulate, and resolve project challenges	Employability (Understanding / Analysing)	→ **Associated Activities:** Problem identification, requirement analysis, feasibility study, literature/technology survey, project proposal preparation

	effectively.		→ **Assessment Methods:** Project proposal review, mentor evaluation, oral questioning
CO2	Design and develop a system/interface/database covering all modules of the project.	Skill Development (Designing / Applying)	→ **Associated Activities:** System architecture design, UI/interface design, database schema design, module-wise implementation → **Assessment Methods:** Design review, phase-wise project evaluation, code walkthrough
CO3	Apply appropriate tools and techniques for testing and implement project modules with a consolidated project report.	Skill Development (Applying / Creating)	→ **Associated Activities:** Test case design, functional testing, debugging, integration of modules, preparation of technical documentation → **Assessment Methods:** Test case evaluation, project execution review, report assessment
CO4	Demonstrate the working project to the end user along with system and user documentation.	Entrepreneurship (Evaluating / Communicating)	→ **Associated Activities:** Project demonstration, user interaction simulation, preparation of user manual, presentation of outcomes → **Assessment Methods:** Final project demo, viva-voce, user manual and presentation evaluation

Prerequisites, if any:

1. Basic knowledge of programming concepts.
2. Prior exposure to data structures and algorithms fundamentals.
3. Understanding of software development lifecycle (SDLC) and basic problem-solving techniques.
4. Familiarity with at least one development platform or tool relevant to the project domain (e.g., Android Studio / Web technologies / AI libraries / DBMS tools / Computer Network Tools).

Topics of Course:

Your End-Semester Presentation will be evaluated based on the following criteria:

- **Project Understanding:** Evaluation begins with assessing whether the students have a clear understanding of the project objectives, scope, and requirements. This includes examining the problem statement, objectives, and proposed solutions.
- **Research and Literature Review:** Evaluation involves reviewing the depth and breadth of the literature review conducted by the students. This includes assessing the relevance, currency, and comprehensiveness of the literature surveyed.
- **Methodology and Approach:** Assessing the appropriateness and rigor of the methodology chosen by the students is crucial. This includes evaluating the research design, data collection methods, tools, and techniques utilized.
- **Implementation and Execution:** Evaluators examine the students' ability to implement the proposed solution effectively. This involves assessing the technical skills demonstrated, the robustness of the implementation process, and any challenges encountered.
- **Results and Analysis:** The evaluation includes an analysis of the results obtained from the project. This involves examining the accuracy, validity, and reliability of the findings, as well as the effectiveness of the data analysis techniques employed.
- **Innovation and Creativity:** Assessing the level of innovation and creativity demonstrated by the students in addressing the project objectives is important. This includes evaluating novel approaches, solutions, or insights generated.

- **Documentation and Presentation:** Evaluation encompasses the quality of documentation and presentation of the project. This includes assessing the clarity, organization, and professionalism of the project report, as well as the effectiveness of the oral presentation.
- **Critical Thinking and Problem-Solving Skills:** Evaluators assess the students' ability to critically analyze problems, identify solutions, and make informed decisions throughout the project lifecycle.
- **Teamwork (if applicable):** For group projects, evaluation may include assessing the effectiveness of teamwork, collaboration, and communication among team members.
- **Overall Contribution and Impact:** Finally, evaluation considers the overall contribution of the project to the field of study or industry, as well as its potential impact and relevance.

CO-PO Mapping:

Course (Level) and Course Outcome Mapping to Program Outcomes *[Three Levels of Mapping: 1: means weak, 2: means medium and 3: means strong]*

Course (Level) Mapping PO1		PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	PSO 3	PSO 4
Mapping of Course Outcomes	CO 1	2	3	1	1	2	-	-	1	-	-	-	1	1	1	1	-
	CO 2	2	2	3	-	3	1	1	-	2	-	1	-	2	2	1	1
	CO 3	2	2	2	3	3	-	-	-	1	2	1	1	3	2	2	1
	CO 4	-	1	2	-	1	2	1	3	3	3	2	1	1	1	-	-

Learning Resources:

Self-learning and/or resources as prescribed by the mentor(s).

Experiential Learning: 100%

Assessment Pattern: The final grade will be based on the marks/ grades obtained in the mid- semester and end-semester evaluation and other assessments defined in the assessment table. The relative grading method described in the university's academic regulations will be followed to grade the students. The student must secure a minimum of 40% marks after completing all the assessments in the following table to become eligible for grading.

Component	Weightage	Evaluation Schedule/Timeline	Rubrics/Remark	CO Mapping
Continuous Evaluation	25%	Continuous evaluations conducted as per the instructions of the mentor throughout the semester.	Participation + Demonstration + Viva (conducted by the mentor as per the rubrics)	CO1, CO2, CO3
Mid Semester Examination	25%	Second week Oct of 2026	Assessed by the Mentor / Panel (if required)	CO1, CO2, CO3
Report	10%	Last week of Nov 2026	Detailed report. The format will be shared.	CO1, CO2, CO3. CO4
			Project Presentation and Demonstration.	

End Semester Presentation	40%	Last week of November 2026	The component will be evaluated by a panel of experts.	CO1, CO2, CO3, CO4

Student Responsibilities:

- Actively participate in all project-related sessions, reviews, and evaluations, and follow the prescribed phase-wise development plan.
- Develop projects using approved tools and technologies, ensuring functional correctness, usability, and adherence to course guidelines provided by the mentors and coordinators.
- Submit all deliverables (code, reports, documentation) on time and demonstrate individual understanding and contribution to the mentor.
- Maintain academic integrity by submitting original work and responding constructively to feedback for continuous improvement.

Attendance Policy:

Students are expected to meet the mentor regularly and maintain the prescribed attendance percentage. Failure to do so will be treated as non-compliance with the university attendance policy.

Recourse examination policy:

As per University policy.

Make-up policy:

No make-up work will be given for unexcused absences. The faculty needs to be informed in advance in case the student is not going to be able to submit an assignment or take any evolution component, and it is at the discretion of the faculty to sanction make up for an evaluation component.

Academic dishonesty/cheating/plagiarism:

Plagiarism and dishonesty in any form in any evaluation component will lead to appropriate disciplinary action.